

Thanuish Kumar Sathiaraj

COMPUTER VISION & SOFTWARE DEVELOPMENT | PHYSICAL AI | AI/LLM SYSTEMS

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M.Sc. Computer Science student specializing in computer vision and moving toward Physical AI: perception systems that understand the real world, combined with large language models. At the Max Planck Institute I build 3D human-body reconstruction pipelines, operate multi-camera and 4D capture hardware, and develop the backend software that runs it in production. Alongside that I build many projects on my own, purely to learn and understand things properly, from deepfake detection and object detection to agentic AI applications using LangGraph and the Model Context Protocol (MCP), including building my own MCP servers.

EDUCATION

M.Sc. Computer Science, University of Stuttgart, Germany | Apr 2025 – Apr 2028 (expected) | Coursework: Computer Vision, Reinforcement Learning, Knowledge Graphs

B.E. Computer Science & Engineering, Anna University | Aug 2020 – Jun 2024 | Coursework: Machine Learning & AI, DBMS, Operating Systems | Languages: English C1, German A2 (studying)

PROFESSIONAL EXPERIENCE

Student Research Assistant, Computer Vision & Software Engineering

Max Planck Institute for Intelligent Systems (Perceiving Systems Dept.), Tübingen, Germany | Nov 2025 – Apr 2026

- **3D reconstruction:** built computer vision pipelines and a real-time viewer for 3D human-body reconstruction from synchronised multi-camera video, using SMPL-X parametric body models with anatomical keypoint tracking and multi-perspective 3D representation, supporting PhD research on MAMMA (Markerless Accurate Multi-Person Motion Acquisition).
- **Perception hardware:** calibrated and operated a Vicon Nexus multi-camera motion-capture system and 3D and 4D full-body scanners through complex dynamic capture sessions, working with real perception hardware end to end.
- **Model serving & backend:** developed backend services and internal web tooling with Python and the Flask framework, exposing pipelines through REST APIs and integrating models into a live internal service so newly uploaded data is processed and reflected automatically.
- **Data pipelines & database:** built and maintained ETL pipelines that collected, cleaned and validated large 3D capture datasets into structured, queryable form, with database development and output visualization.
- **Deployment:** ran the system as containerized Docker services on Linux, staged across 2D, 3D and visualization components so a fault isolates to a single stage.
- **Evaluation & QA:** reviewed model outputs, separated successful reconstructions from failures, diagnosed root causes, and fed concrete improvement proposals back to researchers for the next iteration.
- **Engineering process:** Agile/Scrum, Jira, Git and GitHub with branching, pull requests and code review, shared technical documentation. Referred by Tsvetelina Alexiadis, Data Team Lead.

Machine Learning Engineer Intern

Shiash Info-Solutions | Apr 2024 – Feb 2025

- Built an end-to-end CNN image-classification system in Python, owning the full lifecycle from data collection and cleaning through model training, validation and deployment.
- Improved the model with Bayesian hyperparameter optimization, data augmentation, normalization and dropout regularization, and built automated preprocessing pipelines that cut training time by roughly 20%.
- **Stack:** Python, TensorFlow, Keras, OpenCV, scikit-learn, NumPy, Pandas.

PROJECTS

Deepfake Detection in Images and Video (Bachelor's Thesis)

Python, TensorFlow, Keras, OpenCV, MTCNN, Inception-ResNet, LSTM, Grad-CAM, SHAP

- Built, as sole developer, a cascading computer vision pipeline for detecting manipulated images and video: MTCNN face detection, Inception-ResNet CNN feature extraction, classification, and an LSTM stage to use temporal information across video frames.
- Reached 91.7% accuracy and F1 0.89, a 12.4% improvement over standalone models, trained on 140GB+ of image and video data on Linux with preprocessing runtime cut by 35%.
- Applied Grad-CAM and SHAP to interpret model decisions and diagnose failure cases, validated with 5-fold cross-validation, because a perception model you cannot explain cannot be trusted.

Helmet Detection & License Plate Recognition (Object Detection + OCR)

Python, YOLOv6, OpenCV, Tesseract/EasyOCR

- Deployed YOLOv6 for multi-class real-time object detection (motorcycle, rider, helmet status) in a live inference pipeline, and built a region-of-interest cropping and OCR stage for license-plate extraction, tuned for robustness across real-world lighting and viewing angles.

AI Autopilot: Semantic Caching & Intelligent LLM Routing Gateway (in development, 2026)

Python, FastAPI, Pydantic v2, Valkey, Sentence-Transformers, Ollama, PostgreSQL, scikit-learn, Streamlit, Docker Compose, pytest

- Building a production-style AI gateway that answers each request by the cheapest sufficient path: semantic cache first (embedding similarity search in Valkey), then a deterministic knowledge base, and only then an LLM, so unnecessary inference is avoided entirely.
- Routing layer classifies request complexity with a scikit-learn model and selects a local Ollama model accordingly, behind a provider abstraction; asynchronous quality verification scores each answer and automatically escalates failures to a stronger model, with every request logged to PostgreSQL and visualised in a Streamlit dashboard.

Market Sentinel: Multi-Agent AI System with MCP and RAG

Python, LangGraph, MCP, RAG, Pinecone, pytest, Git | github.com/Thanuish/Market_sentinal

- Built a multi-agent AI system in LangGraph where the final action is structurally unreachable until an evaluator agent writes approval into graph state, exposed the deterministic logic as a Model Context Protocol (MCP) server tool, and added RAG over a Pinecone vector database, backed by a 12-test pytest suite.

VisionARyMenu: AR Application with Visual SLAM

Unity, C#, ARKit, ARCore, AR Foundation, ZXing.Net, NUnit

- Built an augmented-reality application on ARKit and ARCore visual SLAM for plane detection and world tracking, with QR decoding from live camera frames driving remote JSON content retrieval, verified on physical devices.

SKILLS

Computer Vision & 3D: 3D human-body reconstruction, SMPL-X parametric body models, anatomical keypoint tracking, multi-camera systems, 3D and 4D scanning, motion capture (Vicon Nexus), camera calibration, point-cloud-style 3D data, visual SLAM (ARKit/ARCore), object detection (YOLOv6), OCR, MTCNN, OpenCV, image and video processing

Deep Learning & ML: Python, PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, Hugging Face; CNN, LSTM, Inception-ResNet, cascading and ensemble models, hyperparameter optimization, data augmentation, fine-tuning, model evaluation (accuracy, F1, ROC/AUC); interpretability with Grad-CAM and SHAP; Vision Transformers (ViT) from study

LLMs & Agentic AI: LangGraph multi-agent orchestration, MCP (Model Context Protocol) server design and development, RAG with Pinecone vector database, semantic caching and embedding similarity search, LLM routing and evaluation, prompt engineering, LLM APIs (OpenAI/ChatGPT, Anthropic), local model serving with Ollama, fine-tuning T5/FLAN-T5 and LLaMA 3, Sentence-Transformers

Software & Infrastructure: Python (primary), C, C++ (ramp-up), C#, SQL, JavaScript, Bash; Flask and FastAPI, REST API design, Pydantic validation, React, HTML/CSS, ETL and data pipelines, PostgreSQL/SQLite, Valkey/Redis; Docker and Docker Compose on Linux, Git/GitHub with pull requests and code review, Jira, Agile/Scrum, pytest, technical documentation; building CI/CD pipelines